

Mechanistic Interpretability for AI Safety

A Review

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Abstract

Understanding AI systems’ inner workings is critical for ensuring value alignment and safety. This review explores mechanistic interpretability: reverse engineering the computational mechanisms and representations learned by neural networks into human-understandable algorithms and concepts to provide a granular, causal understanding. We establish foundational concepts such as features encoding knowledge within neural activations and hypotheses about their representation and computation. We survey methodologies for causally dissecting model behaviors and assess the relevance of mechanistic interpretability to AI safety. We examine benefits in understanding, control, alignment, and risks such as capability gains and dual-use concerns. We investigate challenges surrounding scalability, automation, and comprehensive interpretation. We advocate for clarifying concepts, setting standards, and scaling techniques to handle complex models and behaviors and expand to domains such as vision and reinforcement learning. Mechanistic interpretability could help prevent catastrophic outcomes as AI systems become more powerful and inscrutable. For an HTML version of the paper, visit <https://leonardbereska.github.io/blog/2024/mechinterpreview/>.

1 Introduction

As AI systems rapidly become more sophisticated and general (Bubeck et al., 2023; Bengio et al., 2023), advancing our understanding of these systems is crucial to ensure their alignment (Ji et al., 2024) with human values and avoid catastrophic outcomes (Hendrycks et al., 2023; Hendrycks & Mazeika, 2022). The field of interpretability aims to demystify the internal processes of AI models, moving beyond evaluating performance alone. This review focuses on mechanistic interpretability, an emerging approach within the broader interpretability landscape that strives to comprehensively specify the computations underlying deep neural networks. We emphasize that understanding and interpreting these complex systems is not merely an academic endeavor – *it’s a societal imperative to ensure AI remains trustworthy and beneficial*.

The interpretability landscape is undergoing a paradigm shift akin to the evolution from behaviorism to cognitive neuroscience in psychology. Historically, lacking tools for introspection, psychology treated the mind as a black box, focusing solely on observable behaviors. Similarly, interpretability has predominantly relied on black-box techniques (Casper et al., 2024), analyzing models based on input-output relationships or using attribution methods that, while probing deeper, still neglect the model’s internal architecture. However, just as advancements in neuroscience allowed for a deeper understanding of internal cognitive processes, the field of interpretability is now moving towards a more granular approach. This shift from surface-level analysis to a focus on the internal mechanics of deep neural networks characterizes the transition towards inner interpretability (Räuker et al., 2023).

Mechanistic interpretability, as an approach to inner interpretability, aims to completely specify a neural network’s computation, potentially in a format as explicit as pseudocode (also called *reverse engineering*), striving for a granular and precise understanding of model behavior. It distinguishes itself primarily through its *ambition* for comprehensive reverse engineering and its strong *motivation* towards AI safety. Our review

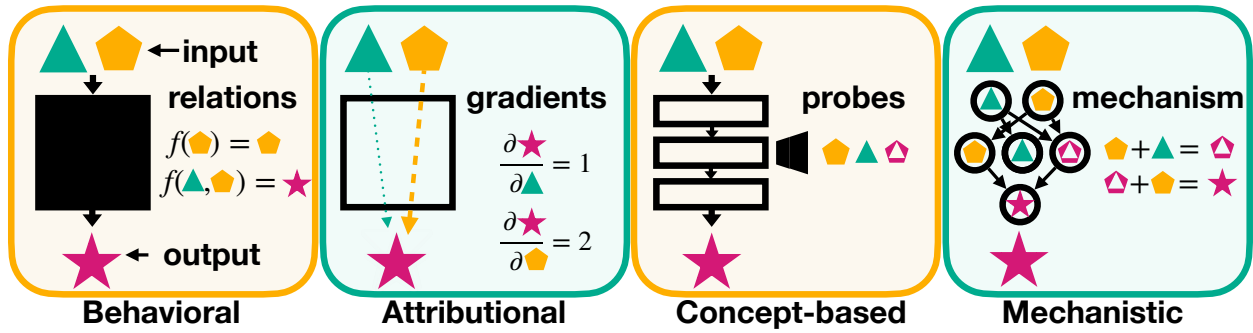


Figure 1: Interpretability paradigms offer distinct lenses for understanding neural networks: **Behavioral** analyzes input-output relations; **Attributional** quantifies individual input feature influences; **Concept-based** identifies high-level representations governing behavior; **Mechanistic** uncovers precise causal mechanisms from inputs to outputs.

serves as the first comprehensive exploration of mechanistic interpretability research, with the most accessible introductions currently scattered in a blog or list format (Olah, 2022; Nanda, 2022d; Olah et al., 2020; Sharkey et al., 2022a; Olah et al., 2018; Nanda, 2023f; 2024). Concurrently, Ferrando et al. (2024) and Rai et al. (2024) have also contributed valuable reviews giving concise, technical introductions to mechanistic interpretability in transformer-based language models. Our work complements these efforts by synthesizing the research (addressing the "research debt" (Olah & Carter, 2017)) and providing a structured, accessible, and comprehensive introduction for AI researchers and practitioners.

The structure of this paper provides a cohesive overview of mechanistic interpretability, situating the mechanistic approach in the broader interpretability landscape (Section 2), presenting core concepts and hypotheses (Section 3), explaining methods and techniques (Section 4), presenting a taxonomy and survey of the current field (Section 5), exploring relevance to AI safety (Section 6), and addressing challenges (Section 7) and future directions (Section 8).

2 Interpretability Paradigms from the Outside In

We encounter a spectrum of interpretability paradigms for decoding AI systems' decision-making, ranging from external black-box techniques to internal analyses. We contrast these paradigms with mechanistic interpretability, highlighting its distinct causal bottom-up perspective within the broader interpretability landscape (see Figure 1).

Behavioral interpretability treats the model as a black box, analyzing input-output relations. Techniques such as minimal pair analysis (Warstadt et al., 2020), sensitivity and perturbation analysis (Casalicchio et al., 2018) examine input-output relations to assess the model's robustness and variable dependencies (Shapley, 1988; Ribeiro et al., 2016; Covert et al., 2021). Its *model-agnostic* nature is practical for complex or proprietary models but lacks insight into internal decision processes and causal depth (Jumelet, 2023).

Attributional interpretability aims to explain outputs by tracing predictions to individual input contributions using gradients. Raw gradients can be discontinuous or sensitive to slight perturbations. Therefore, techniques such as SmoothGrad (Smilkov et al., 2017) and Integrated Gradients (Sundararajan et al., 2017) average across gradients. Other popular techniques are layer-wise relevance propagation (Bach et al., 2015), DeepLIFT (Shrikumar et al., 2017), or GradCAM (Selvaraju et al., 2016). Attribution enhances transparency by showing input feature influence without requiring an understanding of the internal structure, enabling decision validation, compliance, and trust while serving as a bias detection tool, but also has fundamental limitations (Bilodeau et al., 2024).

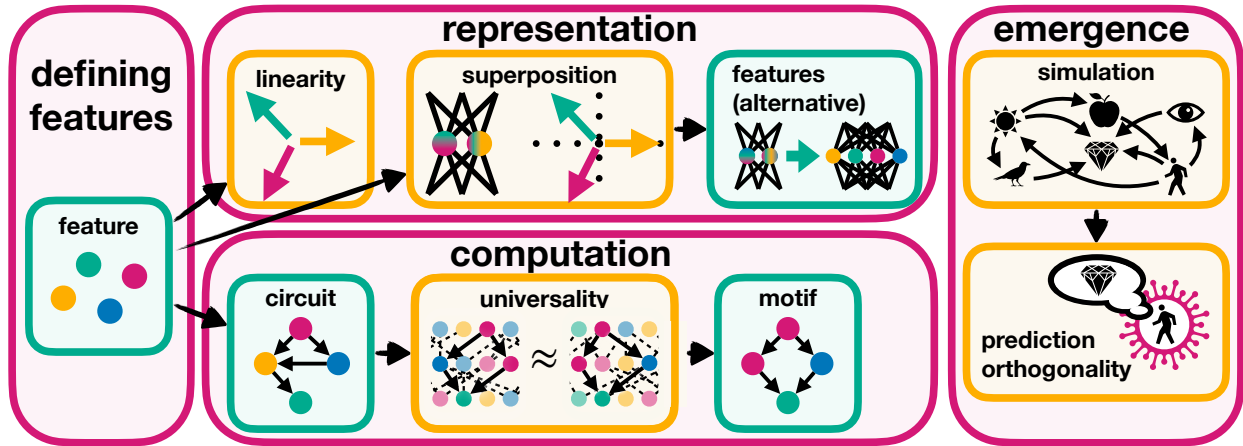


Figure 2: Overview of key concepts and hypotheses in mechanistic interpretability, organized into four subsection (pink boxes): defining features (Section 3.1), representation (Section 3.2), computation (Section 3.3), and emergence (Section 3.4). In turquoise, it highlights definitions like *features*, *circuits*, and *motifs*, and in orange, it highlights hypotheses like *linear representation*, *superposition*, *universality*, *simulation*, and *prediction orthogonality*. Arrows show relationships, e.g., superposition enabling an alternative feature definition or universality connecting circuits and motifs.

Concept-based interpretability adopts a top-down approach to unraveling a model’s decision-making processes by probing its learned representations for high-level concepts and patterns governing behavior. Techniques include training supervised auxiliary classifiers (Belinkov, 2021), employing unsupervised contrastive and structured probes (see Section 4.2) to explore latent knowledge (Burns et al., 2023), and using neural representation analysis to quantify the representational similarities between the internal representations learned by different neural networks (Kornblith et al., 2019; Bansal et al., 2021). Beyond observational analysis, concept-based interpretability can enable manipulation of these representations – also called *representation engineering* (Zou et al., 2023) – potentially enhancing safety by upregulating concepts such as honesty, harmlessness, and morality.

Mechanistic interpretability is a bottom-up approach that studies the fundamental components of models through granular analysis of features, neurons, layers, and connections, offering an intimate view of operational mechanics. Unlike concept-based interpretability, it aims to uncover causal relationships and precise computations transforming inputs into outputs, often identifying specific neural circuits driving behavior. This *reverse engineering* approach draws from interdisciplinary fields like physics, neuroscience, and systems biology to guide the development of transparent, value-aligned AI systems. Mechanistic interpretability is the primary focus of this review.

3 Core Concepts and Assumptions

This section introduces the key concepts and hypotheses of mechanistic interpretability, as summarized in Figure 2. We start by defining features as the basic units of representation (Section 3.1). We then examine the nature of these features, including the challenges posed by polysemantic neurons and the implications of the superposition and linear representation hypotheses (Section 3.2). Next, we explore computation through circuits and motifs, considering the universality hypothesis (Section 3.3). Finally, we discuss the implications for understanding emergent properties, such as internal world models and simulated agents with potentially misaligned objectives (Section 3.4).